

# AJAY ARAVIND PRAKASH

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Full Stack Software Engineer with 4 years building Angular/React frontends and Spring Boot microservices for enterprise telecom platforms. Designs event-driven distributed systems, real-time geospatial dashboards, and AI-powered RAG pipelines serving 8,000+ daily users. Improved NPS by 18%, cut deployment time by 87%, and eliminated 90% of pre-release defects on a platform monitoring 40M+ cable modems.

## EDUCATION

### University of Wisconsin-Milwaukee

May 2027

Master of Science in Computer Science | Machine Learning, NLP, Image Processing, Intelligent User Interfaces

### KPR Institute of Engineering and Technology, Anna University

Apr 2022

Bachelor of Engineering in Computer Science

Undergraduate Research Assistant — AI Lab (Nov 2019 – Dec 2021): Image classification and pattern recognition research.

## PROFESSIONAL EXPERIENCE

### Associate Engineer (Software Engineer) | Virtusa Corporation

Dec 2021 – Aug 2025

- Boosted network monitoring accuracy by 90% and cut issue identification time by 64% for 8,000+ users via ArcGIS.
- Designed 10+ Spring Boot microservices with 99.9% uptime, processing 1M+ daily data records and 200K+ API requests/day.
- Built REST APIs and GraphQL services with Spring Security, JWT, and RBAC for enterprise-grade access control.
- Implemented Kafka event-driven architecture handling 50K+ messages/day, reducing alert latency by 40%.
- Eliminated 850+ UI bugs and improved app load time to 100% by adopting RxState and fixing memory leaks.
- Built a micro frontend Angular monorepo using NX with role-based plugin architecture across 7 modules for 8,000+ users.
- Built 23 real-time D3.js dashboards, accelerating outage detection by 100% across 7 network operations teams.
- Optimized queries across MySQL, PostgreSQL, Oracle, Elasticsearch, and Druid via indexing on high-traffic systems.
- Integrated OpenAI APIs and LangChain RAG pipelines, cutting internal knowledge search time by 60%.
- Automated E2E testing with Cypress and achieved 80% unit test coverage with Jest, eliminating 90% of pre-release defects.
- Reduced deployment time by 87% (2 hrs to 15 mins) by automating CI/CD in Concourse across 4 environments.
- Drove 18% NPS growth over 2 years by delivering mobile device tracking features improving workflow for 8,000+ technicians.

## PROJECTS

### FAFF — Medical Image Enhancement | Python, Streamlit, OpenCV, NumPy, SciPy, scikit-image, FFT, SSIM

- Improved X-ray, MRI, and ultrasound clarity by 40% SSIM using patch-based adaptive filter with FFT texture classification.
- Cut manual tuning to zero by automating filter weight selection per patch; built Streamlit app for upload and comparison.

### Medical Similar Case Retrieval System | Python, TF-IDF, Weighted MinHash, LSH, REST APIs, MySQL

- Cut clinical note search space by 80% using TF-IDF, MinHash, and LSH bucket grouping for sub-second query response.
- Deployed a privacy-aware REST API enabling real-time retrieval of clinically similar cases at scale.

### Financial Transaction Management System | Angular, AWS Lambda, S3, CloudWatch, Blockchain

- Reduced infrastructure overhead to near zero by architecting a serverless app on AWS Lambda, S3, and CloudWatch.
- Accepted by ICCS Committee for innovation; integrated AI spending insights improving user savings behavior in the app.

### Adaptive Learning Interface — HCI Research Study | React, JavaScript, WCAG, Behavioral Analytics

- Validated 100% improvement in user clarity and task ownership via a blind study of two React interface variants.
- Applied WCAG accessibility across both variants; analyzed task completion time and click behavior across all participants.

## TECHNICAL SKILLS

**Languages:** Java, JavaScript, TypeScript, Python, C, SQL

**Frontend:** Angular, React, HTML5, CSS3, Angular Material, RxState, Redux, D3.js, NX Monorepo, Storybook, Figma

**Backend:** Spring Boot, Spring MVC, REST APIs, GraphQL, Microservices, Node.js, Kafka, Hibernate, OpenAPI/Swagger

**AI / ML:** LangChain, OpenAI APIs, RAG Pipelines, TF-IDF, LSH, MinHash, OpenCV, scikit-image, NumPy, SciPy, FFT

**Databases:** MySQL, PostgreSQL, Oracle PL/SQL, MongoDB, Elasticsearch

**Cloud & DevOps:** AWS (EC2, Lambda, S3, CloudWatch), Docker, Kubernetes, Concourse CI, Terraform, Ansible, Git, CI/CD

**Tools:** Prometheus, Grafana, Postman, Swagger UI, Power BI, JIRA, Confluence, Jest, Cypress, JUnit, LoadRunner

## CERTIFICATIONS & ACHIEVEMENTS

OCJA Java 8 — Oracle | AWS Developer Associate | AWS Solution Architect | Google Cloud Engineering Specialization

4th Place & Best Presentation — National Hackathon, Manipal University (2019) | 70% Sponsorship Growth, Ignitreron Techno-Fest (2021–22)